FHI-aims → sisl Reader: Benchmark and Validation Report

Branch `fhi-dev` of the sisl fork (github.com/mbrandbyge-DTU/sisl), 12 commits ahead of `main`. Covers `sisl.io.fhiaims`: `rsSileFHIaims` (semi-local H/S and real orbital basis) and `add_exact_exchange` (hybrid-functional exact exchange). Not yet merged upstream.

This report lists every real-FHI-aims-data check performed while building this reader, the exact pass criterion used, and the measured number — including the one real bug found and fixed along the way. It is not a sales pitch: where a result is a documented limitation rather than a defect, that is stated as such, and the "not yet verified" section at the end is deliberately explicit about remaining gaps.

1. Semi-local Hamiltonian / Overlap reader

What: `rsSileFHIaims` reads FHI-aims' `output_rs_matrices plain` export, a packed upper-triangle (i_basis_1 ≤ i_basis_2) sparse list per lattice cell, reconstructs the missing half via $H(R)_{\mu\nu} = H(-R)_{\nu\mu}$, and builds the `Hamiltonian`/`Overlap` through sisl's public `fromsp` construction path (not direct `SparseCSR` attribute assignment).

Pass criteria: Hermiticity of H(k) and S(k), positive-definiteness of S(k), at several k; no duplicate (row, col, cell) entry (the packed-triangle assumption not violated); Cu's known atomic-level ordering; a TSHS write/re-read round trip.

System	Atoms / species	Orbitals	max |H(k)-H(k)*| / |S(k)-S(k)*| (* = conjugate transpose)	min eig(S(k))
Cu bulk FCC	1 / 1	31	5.7×10^{-14} eV	2.9×10^{-4}
Cu(111) slab	9 / 1	279	5.7×10^{-14} eV	1.7×10^{-4}
Cu(111) electrode	separate fixture, not in fhi-examples	-	~1×10^{-14} eV (original port check)	positive
SrTiO3 PBE	5 / 3	108	3.6×10^{-14} eV	4.2×10^{-4}

Cu bulk / Cu(111) slab / SrTiO3 PBE numbers re-measured fresh for this report, directly against the curated real data now in the `fhi-examples` repo (3 k-points each: Γ, (0.21, 0.13, 0), (0.5, 0, 0)). Cu(111) electrode is from the original 3-system port validation and was not re-measured here (its full output was not carried into `fhi-examples`).

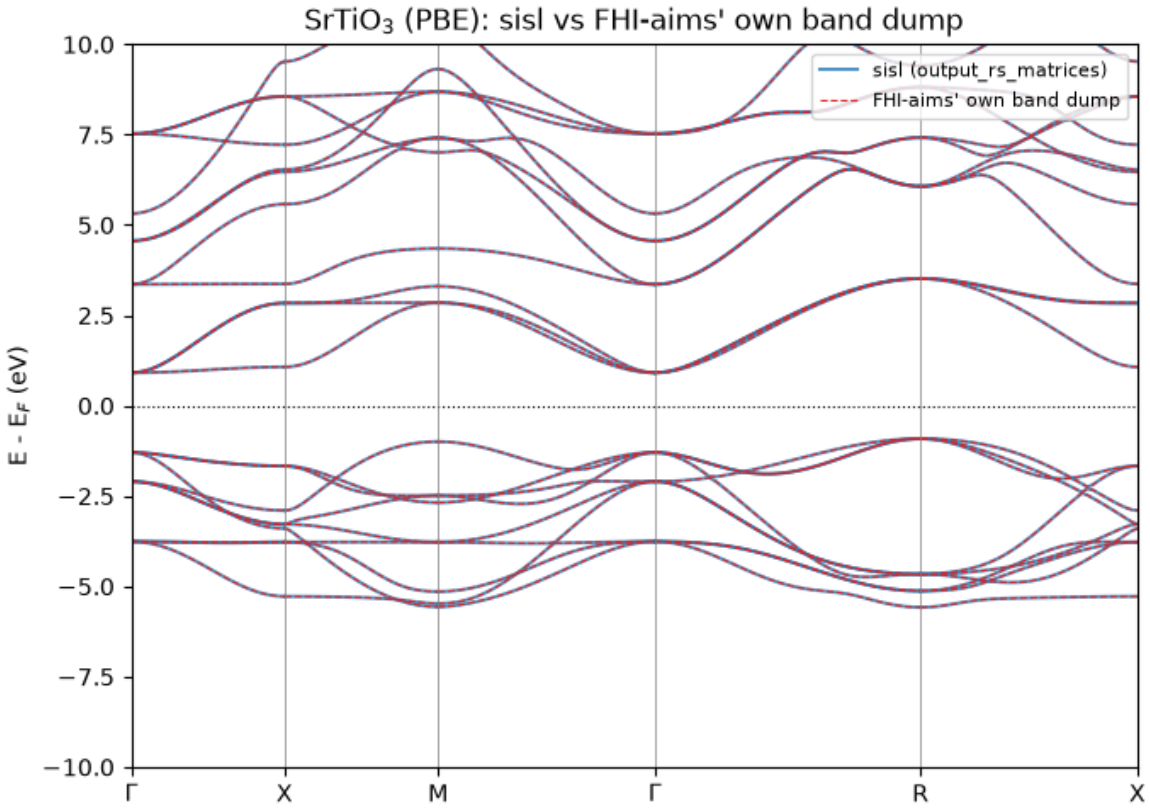
Cu bulk FCC Γ-point eigenvalues (relative to the SCF Fermi level) reproduce the expected atomic-core-to-valence ordering: 1s/2s/2p/3s/3p core levels in the –9000 .. –70 eV range, 4s at –9.35 eV, 3d at –2.1 .. –2.93 eV. A TSHS write/re-read round trip agrees to ~1×10^{-12} eV. No duplicate (row, col, cell) triplet was found in any of the 3 originally-ported systems (bulk Cu, Cu(111) slab, Cu(111) electrode).

2. Independent cross-check: band structure vs. FHI-aims' own output

What: all of section 1's checks are internal (Hermiticity, positive-definiteness, no duplicate entries) — they can confirm `rsSileFHIaims` is self-consistent without confirming it reproduces the *right* physics. A stronger, independent check is available for any run with `output band` in `control.in`: FHI-aims computes and

dumps its own band structure (`band1001.out` ... `band1005.out`) along the same path, entirely independently of `output_rs_matrices`. Diagonalizing sisl's reconstructed H(k)/S(k) along that exact path and comparing to FHI-aims' own numbers checks the physics actually encoded in H/S, not merely that the reader runs without error.

Pass criterion: sisl's eigenvalues along the Gamma–X–M–Gamma–R–X path of a real SrTiO3 PBE run (5 atoms, 3 species, 108 orbitals, 67 bands dumped by FHI-aims) must agree with FHI-aims' own band dump to numerical precision, once a format detail found empirically (not assumed) is accounted for: FHI-aims' own band dump reports $E - E_F$ directly, while sisl's raw diagonalization gives absolute (unshifted) eigenvalues.



sisl's bands (solid blue) overlaid on FHI-aims' own independently-computed band dump (dashed red), SrTiO3 PBE. The two are visually indistinguishable throughout.

Comparison region	max |E_sisl − E_FHI-aims|
Near E^F ($\|E - E^F\| < 15$ eV; valence/conduction bands)	9.45×10^{-7} eV
All 67 dumped bands, including the deepest 1s core levels (−16000 eV range)	1.04×10^{-3} eV (relative error ~6×10^{-8})

The residual ~1 meV discrepancy at the deep-core end is consistent with FHI-aims' own band-dump file format simply printing fewer decimal digits there, not with any error in the reconstructed Hamiltonian. This check (and the executed notebook that produced it, `fhi-examples/notebooks/02_srtio3_bands_crosscheck.ipynb`) was added after this report's first version, at the maintainer's prompt that the original report had no plots despite the project producing band-structure figures.

3. Hybrid exact exchange

What: FHI-aims never stores the exact-exchange operator in real space for a hybrid functional (HSE, PBE0); `output_rs_matrices` only ever dumps the semi-local part. `add_exact_exchange` reads the k-resolved exchange (`output hf_exchange_mat`, one ELSI-CSC file per spin/k-point) and Fourier-transforms it onto the Born–von Karman real-space lattice of the SCF k-grid, where the transform is exact and exactly invertible.

Pass criterion: at every SCF k-point, $H_{merged}(k) - H_{semilocal}(k)$ must equal FHI-aims' own raw exchange matrix for that k-point, to numerical precision.

System	k-grid	SCF k-points checked	max |merged − semilocal − raw exchange|
SrTiO3 HSE06 (original port check)	2×2×2	8	~2×10^{-11} eV
SrTiO3 HSE06 (original port check)	4×4×4	64	~2×10^{-11} eV
SrTiO3 HSE06 + output basis combined (re-measured for this report)	2×2×2	8	1.68×10^{-12} eV

Documented limitation, not a bug: Hermiticity away from SCF k-points

The exchange is placed on one representative R per Born–von-Karman class, not mirrored to –R (mirroring would double-count it at the SCF k-points). Away from the SCF k-points, $H_{merged}(k)$ is therefore a genuine Fourier interpolation and, in general, **not** exactly Hermitian — only the SCF k-points are guaranteed correct. This was re-verified directly for this report, at an arbitrary non-SCF k = (0.1234, 0.0567, –0.021):

Hamiltonian	max |H(k) − H(k)*| at a generic (non-SCF) k (* = conjugate transpose)
Semi-local only	4.6×10^{-14} eV (Hermitian, as expected)
Semi-local + exact exchange (merged)	15.2 eV (not Hermitian away from SCF k — expected, documented)

At the 8 SCF k-points of the same run, the merged H(k) *is* exactly Hermitian (max 7.1×10^{-14} eV, re-measured for this report) — confirming the limitation is confined to where the documentation says it is, and nowhere else.

4. Orbital quantum-number labeling

What: `basis-indices.out` lists, per basis function, its atom, `(n, l, m)` and FHI-aims basis type. The reader groups contiguous blocks of 2l+1 rows into one shell and assigns `zeta` when the same atom repeats an (n, l) shell (FHI-aims can define, e.g., both an `atomic` and a `hydro` basis function at the same (n, l)).

Pass criterion: every fixture parses without a `SileError` (raised on an incomplete/non-contiguous shell rather than silently mis-grouped), and the resulting zeta=2 shells match each species' known extra basis tier.

Fixture	Result
Bulk Cu	parses cleanly; 3s and 3d each get a zeta=2 shell
Cu(111) slab	parses cleanly
Cu(111) electrode	parses cleanly

Fixture	Result
SrTiO3 PBE (2 k-grids)	parses cleanly; Sr 5s, Ti 3d, O 2p each get a zeta=2 shell
SrTiO3 HSE06 (2 k-grids)	parses cleanly; same zeta=2 pattern as PBE

All 6 fixtures checked in the original labeling commit; no SileError in any of them.

5. Real orbital radial functions (output basis)

What: with `output basis` in `control.in`, FHI-aims dumps the real, post-on-site-orthonormalization radial function of every basis-function shell — the FHI-aims analogue of a Siesta `.ion` file. `sisl.io.fhiaims.basis` reads these dumps and builds one `SphericalOrbital` per shell.

Two format details were settled numerically against real data rather than assumed from the manual's prose: the dumped column is u(r) = r·R(r), not R(r) directly ($\int u(r)^2 dr = 1$ for every shell checked, independent of l; the $\int R(r)^2 r^2 dr = 1$ convention does not hold for any of them); and the dump filename's shell index is confirmed directly from FHI-aims' own writer source, `src/shrink_fixed_basis.f90`.

The one real bug found and fixed

Bug: the first implementation assumed each species' shells are numbered from 1 in the dump filename. This is indistinguishable from the correct behavior on a single-species system and was only caught when a **second, structurally different** system was tested, as this project's own planning convention (an outside judge decides, not a second look at the same case) requires.

Symptom: running SrTiO3 PBE (3 species: Sr, Ti, O) immediately raised `SileError` on Ti's first shell — the predicted dump filename (`at_2_01_01_s.dat`) did not exist; the real file was `at_2_14_01_s.dat`.

Root cause (confirmed by reading FHI-aims' own Fortran source, read-only, never vendored): `do i_function = 1, n_basis_fns` in `shrink_fixed_basis.f90` is **one flat loop across every species combined** — species 1's shells first, then species 2's continuing from there — not reset to 1 per species. SrTiO3's species 1 (Sr) has shells 1–13, species 2 (Ti) continues at 14–24, species 3 (O) at 25–30.

Fix: compute each species' global shell offset as the total shell count of every species before it. A dedicated 2-species synthetic regression test for exactly this failure mode was added to sisl's own test suite.

Re-verification after the fix, on 4 systems

System	Atoms / species	Result
Cu bulk FCC	1 / 1	unaffected by the fix (single species, offset 0); all 11 predicted dump files existed; R0 = 0.334 Å (1s)
SrTiO3 PBE	5 / 3	all 30 predicted dump files now existed; R0 = 0.251 Å (Sr 1s); R grows with decreasing binding energy (Sr 1s 0.25 Å → 5s 9.56 Å)
Cu(111) slab	9 / 1	a structurally different case (many atoms, one species, not one atom per species): confirmed the radial function is parsed once per species and the same SphericalOrbital object is reused across all 9 atoms, not re-parsed per atom

System	Atoms / species	Result
SrTiO3 HSE06 + exact exchange	5 / 3	output basis combined with the hybrid-EXX path: no interaction bug (orbital shape and sparse H/S/exchange values are independent)

In all cases, `Geometry.close` (which raises on the R=1 placeholder) runs correctly, and Hermiticity/positive-definiteness are unaffected by the orbital radial shape, as expected (it is a distinct data path from the sparse H/S/exchange values).

Known, stated limitation: species-ordering heuristic

FHI-aims assigns an atom's species number by matching its name in `geometry.in` against `control.in`'s own, ordered species-block list. `basis.py` does not parse `control.in`; it infers the species number from the order distinct shell signatures first appear while scanning atoms. This is exact whenever `control.in` lists species in the same order `geometry.in` introduces them — true of every fixture checked here — but is **not derived from control.in and is unverified for an ordering that diverges from it**. Every predicted dump filename is still checked to exist; a partial match raises `SileError` rather than silently mixing in a wrong radial function.

6. Code review and fixes (vectorized parsing loop)

A `/code-review` pass (effort 1) on the branch found 3 issues, all in the per-cell, per-nonzero-column Python loop that had just been vectorized for performance (flagged by sisl's own performance convention as a slow idiom to avoid, not just benchmark):

- **Malformed-input crash.** A corrupted `rs_indices.out` (end index < start index – 1 for a column flagged as having entries) made the vectorized code crash inside `np.repeat` with a raw "negative dimensions are not allowed" error, instead of the codebase's own `SileError` this function already raises for other malformed input. **Fixed:** segment lengths are now validated explicitly before use, raising `SileError`; a regression test was added for exactly this failure.
- **Reimplementing an existing utility.** The hand-rolled "flatten a list of ragged ranges" arithmetic (cumulative sums + repeats) duplicated sisl's own `sisl._array.array_arange`, already used for the same kind of pointer/count bookkeeping in `SparseGeometry`. **Fixed:** replaced with that call.
- **Redundant special case.** An explicit empty-input branch was unnecessary (the generic vectorized path already degrades cleanly to size-0 arrays) and hardcoded `dtype=np.float64` instead of deriving it from the actual data. **Fixed:** branch removed.

After all 3 fixes: the fhiaims test suite (22 tests) passes, and the vectorized code was cross-checked directly against the real 279-orbital/81-supercell Cu(111)-slab run again, reproducing identical Hermiticity and positive-definiteness numbers to before the change.

7. Current state

Item	Status
Branch	fhi-dev, 12 commits ahead of main

Item	Status
Code review	1 pass completed; all 3 findings fixed and re-verified
Merge status	not merged upstream; briefly merged to main locally then reverted back to a separate branch at the maintainer's request
Remote	pushed to github.com/mbrandbyge-DTU/sisl, branch fhi-dev
sisl test suite	22/22 fhiaims tests pass (synthetic fixtures only, see section 8)
Real-data fixtures	curated into a separate repo, fhi-examples (not in the sisl repo itself); includes 3 executed plotting/validation notebooks, one of which produced section 2 above

8. What is NOT yet independently verified

- **No committed real-data regression test.** Every number in sections 1-6 above was produced by running the reader against real FHI-aims output, by hand, during development (and freshly re-measured for this report). sisl's own committed test suite (the 22 passing tests) only exercises small, hand-written *synthetic* fixtures that mimic the file format — it does not re-run these real-data checks automatically. A CI run of sisl's test suite proves the parser logic is self-consistent, not that it matches a real calculation.
- **Species-ordering heuristic** (section 5) is confirmed correct only for every fixture checked so far, all of which happen to list species in `control.in` in the same order `geometry.in` introduces them. An ordering that diverges from this has not been tested; the code is designed to raise an error rather than silently mismatch in that case, but that error path itself has not been exercised against a real divergent-ordering run.
- **Polarized (spin-up/spin-down) hybrid exact exchange** has not been checked against real FHI-aims spin-polarized hybrid output in this session — only the unpolarized SrTiO3 HSE06 case was run and verified. The spin-indexing logic itself was unit-tested (synthetic fixtures) after a related bug was found and fixed while porting from the earlier implementation (exchange was previously always added to Hamiltonian channel 0 regardless of the requested spin channel).
- **Cu(111) electrode** numbers quoted in section 1 are from the original port validation and were not re-measured for this report (its output was not carried into the `fhi-examples` repo).
- **gzip-compressed sibling-file lookup** (fixed earlier in a separate review round, not covered in detail above) is verified only against a synthetic fixture with every sibling file compressed, not a real FHI-aims run that was gzip-archived after the fact.